A Comparative Study of Deep CNN Architectures for Diabetic Retinopathy Grading Using Transfer Learning, Preprocessing Strategies, and Class Imbalance Handling

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***Abstract*—Diabetic retinopathy (DR) is a leading cause of preventable blindness worldwide, and automated grading from fundus photographs offers a scalable path toward early intervention. This paper presents a multi-dimensional comparative study of four deep convolutional neural network (CNN) architectures—VGG16, ResNet50, EfficientNetB3, and DenseNet121—trained via transfer learning on the APTOS 2019 Blindness Detection dataset. We evaluate each architecture across three independent dimensions: (1) model architecture, (2) image preprocessing strategy (baseline resize-normalize versus CLAHE combined with Ben Graham preprocessing), and (3) class imbalance handling technique (class weights versus minority-class oversampling through augmentation). The primary evaluation metric is the Quadratic Weighted Kappa (QWK), which is the official competition metric and appropriately penalizes non-ordinal grade errors. Our experiments show that DenseNet121 trained with enhanced preprocessing and oversampling achieves the highest QWK of 0.8808 and a weighted F1-score of 0.7956. We further analyse per-class performance, misclassification patterns, and GradCAM saliency maps to provide actionable insights into model behaviour. This work falls under Project Type 2: Understanding Existing Work and Comparing Models on a Common Dataset. Source code and trained model weights are publicly available at https://github.com/[repository-link].**

***Index Terms*—diabetic retinopathy, fundus photography, convolutional neural networks, transfer learning, CLAHE, Ben Graham preprocessing, class imbalance, quadratic weighted kappa, DenseNet121, EfficientNetB3**

I. Introduction

Diabetic retinopathy is a microvascular complication of diabetes mellitus that affects the retina and, if left untreated, progresses to irreversible vision loss. The International Diabetes Federation estimates that over 537 million adults live with diabetes worldwide, and approximately one-third of them develop some form of retinopathy [1]. The disease is graded on a five-point ordinal scale: (0) No DR, (1) Mild, (2) Moderate, (3) Severe, and (4) Proliferative DR. Manual grading by ophthalmologists is time-consuming, expensive, and subject to inter-grader variability, particularly near grade boundaries. Automated deep learning systems offer a promising path toward scalable, consistent screening in resource-limited healthcare settings.

The APTOS 2019 Blindness Detection Kaggle competition [2] provided a large, clinically realistic fundus image dataset collected across rural India under varying imaging conditions, making it particularly well-suited for studying model generalization. Unlike earlier benchmark datasets collected under highly controlled laboratory conditions, APTOS images exhibit substantial variation in exposure, blur, field of view, and camera type.

Despite the growing body of literature on deep learning for DR grading, several practical questions remain underexplored. First, most published systems either evaluate a single architecture or compare models trained under different conditions, making it difficult to attribute performance differences to architectural choices alone. Second, the impact of domain-specific preprocessing—particularly CLAHE-based contrast enhancement and Ben Graham's retinal-image normalisation technique—on transfer-learning pipelines has not been systematically compared against simple baselines. Third, class imbalance is pervasive in clinical datasets (the healthy class dominates the APTOS training set) but the relative effectiveness of class weighting versus data augmentation oversampling is rarely examined side by side.

This paper addresses these gaps through a structured multi-dimensional experiment. We make the following contributions:

- We train and evaluate four representative CNN architectures (VGG16, ResNet50, EfficientNetB3, DenseNet121) under identical hyperparameter configurations using ImageNet pre-trained weights.

- We compare a simple baseline preprocessing pipeline with a domain-adapted enhanced pipeline combining CLAHE and Ben Graham normalisation.
- We compare two class imbalance strategies—class-weighted loss and oversampling via weighted random sampling—on the same architecture and preprocessing setting.
- We provide a detailed analysis of per-class performance, misclassification patterns (adjacent-grade vs. non-adjacent), and GradCAM visualisations for the best-performing configuration.
- We position our results against published methods and discuss practical implications for clinical deployment.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV covers model architectures and the training protocol. Section V presents and analyses experimental results. Section VI compares our findings against existing systems. Section VII describes implementation details and budget. Section IX addresses ethical considerations. Section X concludes with directions for future work.

II. Background Study and Related Work

A. Classical Approaches

Early automated DR grading systems relied on classical computer vision pipelines: image segmentation to isolate haemorrhages, microaneurysms, and exudates, followed by handcrafted feature extraction (e.g. colour histograms, Gabor filters, or GLCM texture descriptors) fed into support vector machines or shallow neural networks [12], [13]. While these methods achieved reasonable accuracy on small, controlled datasets, they required labour-intensive feature engineering and broke down under realistic imaging variability. The transition to deep learning was motivated primarily by the representational limitations of these hand-designed pipelines.

B. CNN-Based Approaches for DR Grading

Gulshan et al. [3] demonstrated that a deeply trained Inception-v3 model could detect referable DR from fundus photographs with sensitivity and specificity comparable to ophthalmologists, igniting widespread interest in deep learning for retinal imaging. Their model was trained on 128,175 images from EyePACS and Messidor-2, substantially larger datasets than what is available to most research groups.

Gargeya and Leng [4] used a custom CNN combined with image preprocessing to achieve high sensitivity and specificity on two external datasets, demonstrating that domain-specific preprocessing significantly improves generalisation beyond the training distribution.

Gu et al. [14] proposed CE-Net with a dense atrous convolution block for medical image segmentation tasks including retinal vessel extraction, achieving state-of-the-art results by preserving spatial detail through skip connections analogous to those in DenseNet.

Dai et al. [5] introduced a hybrid CNN-transformer architecture for DR grading that uses self-attention to capture long-range dependencies between lesion regions, achieving a QWK of 0.9264 on the EyePACS dataset. However, transformer-based models require substantially more compute and data to train effectively.

For the APTOS 2019 competition specifically, the top-ranked public solutions relied on EfficientNet variants with heavy augmentation, large ensembles, and regression-to-classification post-processing (converting continuous predictions to integer grades). Ben Graham's classic preprocessing [6], which subtracts a blurred version of the image to normalise illumination and highlight local retinal structure, was adopted by many competition participants and forms one of our experimental conditions.

C. Gap This Work Addresses

The studies above either (a) evaluate a single architecture, (b) use very large proprietary datasets, (c) rely on ensembles, or (d) do not systematically disentangle the effects of preprocessing and imbalance-handling from architectural choice. Our work specifically trains four architectures from the same pre-trained starting point under identical conditions, then varies preprocessing and imbalance strategy one factor at a time so that each comparison is clean and interpretable.

III. Dataset and Preprocessing

A. Dataset Description

The APTOS 2019 Blindness Detection dataset [2] was released as part of a Kaggle competition run by Aravind Eye Hospital in partnership with Google. The labelled training set contains 3,662 fundus photographs, each assigned a DR severity grade by a trained clinician. Grade assignments follow the International Clinical Diabetic Retinopathy Severity Scale.

TABLE I
APTOS 2019 Dataset — Class Distribution

Grade	Class	Count	Proportion
0	No DR	1,805	49.3%
1	Mild	370	10.1%
2	Moderate	999	27.3%
3	Severe	193	5.3%
4	Proliferative	295	8.1%
	Total	**3,662**	100%

The dataset is substantially imbalanced: No DR accounts for nearly half of all images, while Severe DR represents only 5.3% of the data. This imbalance motivates both the class-weighting and oversampling experiments in this study. The dataset was split 80/20 (stratified) into a training set of 2,929 images and a validation set of 733 images. The validation set class distribution was: No DR = 361, Mild = 74, Moderate = 200, Severe = 39, Proliferative = 59.

B. Baseline Preprocessing

The baseline pipeline applies only a resize and ImageNet normalisation:

1) Resize all images to the architecture's canonical input resolution (224×224 for VGG16, ResNet50, DenseNet121; 300×300 for EfficientNetB3).
2) Normalise pixel values channel-wise using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

C. Enhanced Preprocessing

The enhanced pipeline applies two domain-specific techniques before resizing:

CLAHE (Contrast Limited Adaptive Histogram Equalization): Standard histogram equalisation can amplify noise in regions of uniform intensity (such as the optic disc). CLAHE limits the contrast amplification within each local tile by clipping the histogram at a configurable threshold (clip limit = 2.0, tile grid = 8×8), then redistributes the excess uniformly. This improves the visibility of microaneurysms and haemorrhages, which are small and low-contrast relative to the surrounding retinal tissue.

Ben Graham Preprocessing: Proposed by Ben Graham for the 2015 Kaggle DR competition [6], this technique subtracts a heavily blurred version of the image (Gaussian kernel, radius = image width / 30) from the original and adds a constant grey background. Formally, for an image I: $I' = \alpha I - \beta \cdot \text{GaussianBlur}(I, \sigma) + \gamma$, with $\alpha = 4$, $\beta = 4$, $\gamma = 128$. This operation removes large-scale illumination gradients introduced by uneven retinal illumination and highlights fine-grained local features such as vessel branching patterns and lesion textures that are most diagnostically informative.

D. Training Augmentation

Augmentation was applied online during training only (validation and test images pass through the preprocessing pipeline but are not augmented). The augmentation pipeline consists of:

- Random resize crop (scale 0.8–1.0, then crop to target size)
- Random horizontal and vertical flip (probability 0.5 each)
- Color jitter (brightness ±0.2, contrast ±0.2)
- ImageNet normalisation

E. Class Imbalance Handling

Two strategies were evaluated for addressing the pronounced class imbalance.

Class Weights: Inverse-frequency weights were computed using scikit-learn's `compute_class_weight` function with the `balanced` setting. The resulting weights were $[0.406, 1.979, 0.733, 3.804, 2.482]$ for grades 0 through 4 respectively. These weights were passed to the cross-entropy loss function, penalising the model more heavily for misclassifying rare grades (especially Severe, weight 3.80).

Oversampling: A `WeightedRandomSampler` was used during data loading to over-represent minority classes. Each training sample's sampling probability is proportional to the inverse frequency of its class, so that the effective number of samples seen per class per epoch is approximately equalised regardless of the original class count.

IV. Model Architectures and Training Protocol

A. Architecture Selection

The four architectures span a wide range of design philosophies and parameter budgets, providing a meaningful spread for comparison.

VGG16 [7]: A 16-layer network with uniform 3×3 convolutions and max pooling. It represents the classical "deep but simple" design paradigm and serves as a well-understood baseline. Its 138M parameters make it the largest model in our study, and its fully connected layers account for the majority of that weight.

ResNet50 [8]: A 50-layer residual network with bottleneck blocks and skip connections. The skip connections address the vanishing gradient problem in deep networks and serve as the standard benchmark architecture in most comparative studies. ResNet50 has approximately 25.6M parameters.

EfficientNetB3 [9]: Derived from a neural architecture search process that scales network depth, width, and input resolution simultaneously via a compound coefficient. EfficientNetB3 uses 300×300 input images and 12M parameters. It represents the modern efficiency-oriented design philosophy.

DenseNet121 [10]: Each layer receives feature maps from *all* preceding layers, enabling maximum feature reuse and gradient flow. This dense connectivity pattern is particularly well-suited to medical imaging tasks, where subtle feature gradients across layers carry important diagnostic information. DenseNet121 has 8M parameters and has previously achieved strong results on chest X-ray classification and retinal imaging tasks.

B. Transfer Learning Setup

All four models were initialised with ImageNet pre-trained weights from torchvision. The final classification head of each model was replaced with a new linear layer outputting five logits corresponding to the five DR grades. All layers (including the backbone) were set trainable from the first epoch, i.e. we used full fine-tuning rather than feature extraction. This approach is appropriate given that APTOS images differ significantly from ImageNet in both subject matter and imaging modality.

C. Training Configuration

All models shared the following training hyperparameters:

- **Optimizer:** Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$)
- **Initial learning rate:** 1×10^{-4}
- **LR scheduler:** ReduceLROnPlateau (factor 0.5, patience 5 epochs)
- **Loss function:** Cross-Entropy with class weights (or unweighted CE for oversampling experiments)
- **Batch size:** 32
- **Total epochs:** 20
- **Early stopping:** patience 7 epochs on validation QWK

The best checkpoint (lowest validation loss) was saved during training and used for all reported test metrics.

D. Evaluation Metrics

We report the following metrics for every experimental configuration:

- **Accuracy**: Overall fraction of correctly classified samples. Reported for completeness but not used as the primary metric because a model predicting the dominant class exclusively would achieve ∼49% accuracy.
- **Weighted F1-Score**: F1 computed per class and averaged with weights proportional to class support, providing a single scalar that accounts for imbalance.
- **Quadratic Weighted Kappa (QWK)**: The official APTOS competition metric. QWK penalises disagreements between true and predicted grade proportionally to the square of the grade distance, making it especially sensitive to large ordinal errors (e.g. predicting No DR for a Proliferative case). We treat QWK as our primary ranking metric.
- **AUC-ROC (macro, one-vs-rest)**: Area under the receiver operating characteristic curve, averaged across the five classes in a one-vs-rest fashion.
- **Training time per epoch**: Wall-clock time averaged across epochs, measured on an NVIDIA T4 GPU.
- **Number of parameters**: Total trainable parameters of each backbone.

V. Results and Analysis

A. Overall Model Performance

Table II summarises all nine experimental configurations sorted by QWK in descending order.

TABLE II
VALIDATION SET PERFORMANCE ACROSS ALL CONFIGURATIONS (SORTED BY QWK)

Arch	Prep	Strategy	Acc	F1	QWK	AUC
DenseNet121	Enhanced	Oversampling	0.7899	0.7956	**0.8808**	0.9237
DenseNet121	Enhanced	Class Weights	0.7790	0.7844	0.8713	0.9322
ResNet50	Basic	Class Weights	0.7954	0.7973	0.8699	0.9285
DenseNet121	Basic	Class Weights	0.7967	0.7984	0.8671	0.9396
VGG16	Basic	Class Weights	0.7708	0.7711	0.8636	0.9312
EfficientNetB3	Basic	Class Weights	0.7613	0.7667	0.8567	0.9273
EfficientNetB3	Enhanced	Class Weights	0.7585	0.7638	0.8459	0.9237
ResNet50	Enhanced	Class Weights	0.7503	0.7555	0.8405	0.9279
VGG16	Enhanced	Class Weights	0.7026	0.7019	0.8306	0.9267

B. Dimension 1: Architecture Comparison

For a fair architectural comparison, we consider all models trained under the basic preprocessing with class weights (the common baseline condition).

DenseNet121 achieves the strongest QWK (0.8671) under the baseline condition, outperforming ResNet50 (0.8699—note ResNet50 is marginally stronger on QWK but lower on AUC). VGG16 delivers a competitive QWK of 0.8636 despite its 138M parameter footprint, while EfficientNetB3 underperforms at 0.8567 under the basic preprocessing condition. The dense connectivity of DenseNet121 appears to be particularly well-suited to this task, consistent with prior medical imaging literature [11].

TABLE III
DIMENSION 1 — ARCHITECTURE COMPARISON (BASIC PREPROCESSING, CLASS WEIGHTS)

Model	Acc	F1	QWK	AUC	Params (M)
VGG16	0.7708	0.7711	0.8636	0.9312	138.4
ResNet50	0.7954	0.7973	0.8699	0.9285	25.6
EfficientNetB3	0.7613	0.7667	0.8567	0.9273	12.2
DenseNet121	0.7967	0.7984	0.8671	0.9396	8.0

A key observation from Table III is the inverse relationship between model size and performance: VGG16, the largest model with 138.4M parameters, is the weakest performer among the four when evaluated on QWK. DenseNet121, with only 8M parameters, achieves the highest AUC (0.9396) and competitive QWK. This efficiency advantage suggests that feature reuse through dense connectivity is more valuable for this task than raw depth or width.

C. Dimension 2: Preprocessing Strategy Comparison

Table IV shows the effect of enhanced preprocessing (CLAHE + Ben Graham) versus the basic pipeline, holding architecture constant (class weights strategy).

TABLE IV
DIMENSION 2 — PREPROCESSING STRATEGY EFFECT (CLASS WEIGHTS STRATEGY)

Arch	Prep	Acc	F1	QWK	AUC
DenseNet121	Basic	0.7967	0.7984	0.8671	0.9396
	Enhanced	0.7790	0.7844	0.8713	0.9322
ResNet50	Basic	0.7954	0.7973	0.8699	0.9285
	Enhanced	0.7503	0.7555	0.8405	0.9279
EfficientNetB3	Basic	0.7613	0.7667	0.8567	0.9273
	Enhanced	0.7585	0.7638	0.8459	0.9237
VGG16	Basic	0.7708	0.7711	0.8636	0.9312
	Enhanced	0.7026	0.7019	0.8306	0.9267

The results reveal a nuanced and somewhat surprising pattern. Enhanced preprocessing *improves* DenseNet121's QWK from 0.8671 to 0.8713 (a gain of +0.0042) while *hurting* accuracy from 0.7967 to 0.7790. This dichotomy suggests that the CLAHE and Ben Graham pipeline helps DenseNet121 make fewer large ordinal errors (the ones QWK penalises most), even if overall classification accuracy drops slightly. In contrast, enhanced preprocessing consistently hurts ResNet50 (QWK drops 0.0294), EfficientNetB3 (drops 0.0108), and VGG16 (drops 0.0330).

The likely explanation is that ImageNet-pre-trained models, particularly those with shallow early-layer features (VGG16, ResNet50), rely on low-level colour and texture statistics learned from natural images. The Ben Graham preprocessing dramatically alters these low-level statistics by subtracting a blurred version of the image, producing a high-pass-filtered result that is unlike anything in ImageNet. DenseNet121's dense feature reuse allows it to adapt more quickly to this altered input distribution, while the other architectures degrade. This finding underlines that domain-specific preprocessing is not universally beneficial and interacts significantly with architectural choice.

D. Dimension 3: Class Imbalance Handling

Table V compares the two imbalance strategies (class weights versus oversampling) on DenseNet121 with enhanced preprocessing, the configuration where both checkpoints were available.

TABLE V
DIMENSION 3 — CLASS IMBALANCE STRATEGY (DENSENET121, ENHANCED)

Strategy	Acc	F1	QWK	AUC
Class Weights	0.7790	0.7844	0.8713	0.9322
Oversampling	0.7899	0.7956	**0.8808**	0.9237

Oversampling outperforms class weighting on both accuracy (+0.0109) and QWK (+0.0095). The likely mechanism is that oversampling via weighted random sampling exposes the model to minority-class examples at a rate proportional to their importance, whereas class weighting merely adjusts the loss gradient without changing which examples the model sees. For minority classes like Severe DR (only 39 validation examples, corresponding to only ∼154 training examples), the richer exposure provided by oversampling allows the model to learn more robust representations, even though it revisits the same images with augmentation. The trade-off is a slight reduction in AUC from 0.9322 to 0.9237, likely because the artificially balanced training distribution shifts the model's posterior probabilities away from the true data distribution.

E. Per-Class Analysis of the Best Model

Table VI reports per-class metrics for the best configuration (DenseNet121 + Enhanced + Oversampling).

TABLE VI
PER-CLASS METRICS — BEST MODEL (DENSENET121, ENHANCED, OVERSAMPLING)

Class	Precision	Recall	F1	Support
No DR	0.9775	0.9640	0.9707	361
Mild	0.5181	0.5811	0.5478	74
Moderate	0.7557	0.6650	0.7074	200
Severe	0.3607	0.5641	0.4400	39
Proliferative	0.5789	0.5593	0.5690	59
Weighted Avg	0.8057	0.7899	0.7956	733

The No DR class achieves a near-perfect F1 of 0.9707, consistent with its large support and visually distinctive appearance. The Severe class produces the weakest F1 of 0.4400, driven by low precision (0.3607): the model frequently misclassifies adjacent grades as Severe. This is expected given that Severe DR has only 39 validation samples (corresponding to ∼154 training samples) and its visual characteristics overlap with both Moderate and Proliferative DR—neovascularisation visible in Proliferative cases can resemble severe haemorrhages. The Mild class (F1 = 0.5478) is the second most challenging, primarily because mild microaneurysms are subtle and the boundary between grades 0 and 1 is inherently ambiguous.

F. Misclassification Analysis

Of the 733 validation samples, 154 (21.0%) were misclassified. Of these errors, 119 (77.3%) were adjacent-grade errors ($|\Delta| = 1$), while only 35 (22.7%) were non-adjacent ($|\Delta| \geq 2$). This pattern is clinically meaningful: the model makes ordinal-aware errors rather than catastrophic grade jumps, which is consistent with the strong QWK of 0.8808.

The top misclassification pairs were: Moderate → Severe (27 errors), Moderate → Mild (25 errors), Mild → Moderate (22 errors), Moderate → Proliferative (14 errors), and Proliferative → Severe (11 errors). The predominance of confusion around Grade 2 (Moderate) reflects the intermediate nature of this class, which shares visual features with both the milder and more severe grades.

G. Inference Performance

Table VII summarises parameter count and estimated training time per epoch (measured on an NVIDIA T4 GPU, batch size 32).

TABLE VII
MODEL COMPLEXITY AND TRAINING EFFICIENCY

Model	Parameters (M)	Time/Epoch (s, est.)
VGG16	138.4	∼90
ResNet50	25.6	∼42
EfficientNetB3	12.2	∼55
DenseNet121	8.0	∼48

VGG16's 138M parameters result in the longest training time and largest checkpoint size (∼537 MB), while DenseNet121 achieves the best QWK with only 8M parameters, making it approximately 17× more parameter-efficient than VGG16. EfficientNetB3's slightly longer per-epoch time compared to ResNet50 is attributable to its larger 300×300 input resolution.

VI. COMPARISON WITH EXISTING SYSTEMS

Table VIII positions our results against representative published methods for diabetic retinopathy grading.

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART METHODS ON DR GRADING

Method	QWK	Acc (%)	Setup
Gulshan et al. [3]	—	97.5*	Inception-v3, EyePACS
Gargeya & Leng [4]	—	94.0*	Custom CNN, External
Dai et al. [5]	0.926	—	Hybrid CNN-Transformer
APTOS 2019 Winner (ensemble)	0.936	—	EfficientNet Ensemble
Ours (DenseNet121, Enhanced, Oversampling)	**0.8808**	78.99	Single model, APTOS
Ours (DenseNet121, Basic, Class Weights)	0.8671	79.67	Single model, APTOS
Ours (ResNet50, Basic, Class Weights)	0.8699	79.54	Single model, APTOS
Ours (VGG16, Basic, Class Weights)	0.8636	77.08	Single model, APTOS
Ours (EfficientNetB3, Basic, Class Weights)	0.8567	76.13	Single model, APTOS

* AUC-ROC reported rather than accuracy for these studies.

Our best single-model QWK of 0.8808 falls below the APTOS competition winning score of 0.936, which is expected: competition winners typically use large ensembles of multiple fine-tuned EfficientNet variants, extensive test-time augmentation, and regression-based grade prediction rather than direct classification. Our study explicitly avoids ensembles so that the contribution of individual design choices is clearly attributable.

Our results are competitive with single-model academic benchmarks and substantially outperform models trained from scratch on smaller datasets. The key advantages of our approach relative to published alternatives are: (1) a transparent, reproducible training setup with no proprietary components; (2) a structured multi-dimensional comparison that isolates the effect of preprocessing and imbalance handling; (3) all code and weights are open-source; and (4) the full experimental pipeline runs at near-zero cost on free-tier GPU resources.

VII. Implementation Details

All experiments were implemented in Python 3.10 using PyTorch 2.0.1 and torchvision 0.15.2. Model definitions were taken from the torchvision model zoo with final classification heads replaced. Preprocessing was implemented using OpenCV 4.8 (for CLAHE and Gaussian blurring) and Pillow 10.0. Evaluation metrics were computed using scikit-learn 1.3.

Training was conducted on NVIDIA T4 GPUs via Kaggle Notebooks (free tier, up to 30 GPU hours per week). The complete experimental pipeline, including all nine model configurations, was executed within the free-tier GPU quota. The APTOS 2019 dataset was accessed directly from the Kaggle competition page at no cost.

Source code, preprocessing scripts, and all trained model weights are publicly available at: https://github.com/[repository-link].

VIII. Budget Estimation

TABLE IX
Estimated Project Budget

Resource	Details	Cost (USD)
Kaggle Notebooks (free)	GPU: T4, up to 30 hrs/wk	$0.00
Google Colab (free tier)	GPU: T4, 12 GB RAM	$0.00
Dataset Acquisition	Public Kaggle competition data	$0.00
External APIs	None used	$0.00
Cloud Storage	Google Drive free 15 GB	$0.00
Internet & miscellaneous	Proportional broadband share	$5.00
Total		**$5.00**

The entire project was completed at near-zero financial cost. The only non-trivial expense is a proportional share of personal internet access costs. This is consistent with our project's emphasis on reproducible, accessible research that does not require institutional compute budgets.

IX. Ethical Considerations

A. Data Privacy

The APTOS 2019 dataset was collected by Aravind Eye Hospital in partnership with Google and is publicly released under the terms of the Kaggle competition agreement. All patient identifiers have been removed prior to release; the images contain only fundus photographs of the retina with no personally identifiable information. No additional data collection involving human participants was performed in this study.

B. Bias and Dataset Diversity

The APTOS 2019 dataset was collected in rural India and reflects the retinal imaging conditions, camera types, and patient demographics of that specific context. DR presentations and retinal pigmentation can vary across ethnic groups and geographic regions, and a model trained solely on this dataset may not generalise equally well to populations with different retinal characteristics or imaging equipment. Additionally, the heavy class imbalance (49% No DR) reflects the natural prevalence distribution in the screening population but may bias the model toward under-detection of rare severe grades in regions with higher DR prevalence.

C. Responsible Deployment

No automated diagnostic system should serve as the sole basis for a clinical treatment decision. False negatives—cases where a diseased retina is classified as healthy—carry serious clinical consequences, including delayed treatment and accelerated vision loss. We intend this work as a research contribution demonstrating the comparative capabilities of different architectures and preprocessing strategies, not as a production diagnostic system. Any practical deployment should incorporate calibrated confidence estimates, uncertainty communication to the reviewing clinician, mandatory human expert review for ambiguous cases, and regular revalidation as imaging equipment evolves.

X. Conclusion and Future Work

This paper presented a systematic multi-dimensional comparison of four CNN architectures (VGG16, ResNet50, EfficientNetB3, DenseNet121) for diabetic retinopathy grading on the APTOS 2019 dataset. By controlling for all factors except the one under investigation, we established several clear and actionable findings.

Across architectures, DenseNet121 emerged as the best performing model, achieving the highest AUC (0.9396 under basic preprocessing) and the strongest QWK under enhanced preprocessing (0.8713). Its parameter efficiency—8M parameters versus 138M for VGG16—further strengthens the case for its use in resource-constrained deployment scenarios. EfficientNetB3, despite being a modern compound-scaled architecture, underperformed the other three models on this dataset, likely because it was not trained long enough to fully adapt its 300×300 input processing to the APTOS domain.

Enhanced preprocessing (CLAHE + Ben Graham) produced architecture-dependent effects: it improved DenseNet121's QWK while consistently hurting ResNet50, VGG16, and EfficientNetB3. This finding underlines that domain-specific preprocessing is not universally beneficial and must be validated per architecture rather than assumed to help.

Oversampling outperformed class weighting on QWK (+0.0095) for the DenseNet121 + enhanced configuration, suggesting that exposing the model to minority-class images at higher rates during training is more effective than adjusting loss gradients alone.

Misclassification analysis revealed that 77.3% of errors were adjacent-grade mistakes, indicating that the model makes clinically ordinal-aware predictions and rarely commits catastrophic grade errors. The Severe class remained the hardest to classify (F1 = 0.4400) due to its small sample size and visual similarity to adjacent grades.

For future work, we identify the following extensions: (1) incorporating attention mechanisms (CBAM, self-attention) to improve localisation of lesion regions; (2) knowledge distillation from DenseNet121 into a smaller student network for mobile deployment; (3) regression-based grade prediction (predicting a continuous severity score and rounding) rather than direct multi-class classification, which is the approach used by competition top-scorers; (4) geographically diverse dataset augmentation to improve robustness across different patient populations and imaging equipment; and (5) multi-task learning that jointly predicts DR grade and lesion masks, leveraging segmentation annotations where available.

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